

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

QUESTIONS: 30

Time Duration: 30 Minutes

Total Marks:30

Annexure A

MCQ

Code of Conduct:

I Kishore S.P (192511081) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Circle the most appropriate option for each question.

Q1. Which of the following best describes a Logical Agent in AI?

- | | |
|--|---|
| a) An agent that reacts randomly to the environment | c) An agent that only follows pre-defined scripts |
| b) An agent that uses a knowledge base and logical inference to decide actions | d) An agent that learns from rewards |

Q2. In Propositional Logic, ' $P \rightarrow Q$ ' is logically equivalent to:

- | | |
|----------------------|-------------------------------------|
| a) $P \wedge \neg Q$ | c) $\neg Q \rightarrow \neg P$ only |
| b) $\neg P \vee Q$ | d) Both b and c |

Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:

- | | |
|--|-----------------------------------|
| a) A disjunction of conjunctions | c) A conjunction of literals only |
| b) A conjunction of disjunctions (clauses) | d) A sequence of implications |

Q4. In FOL, the sentence $\forall x: \text{Student}(x) \rightarrow \text{Passes}(x)$ means:

- | | |
|--|---|
| a) Some students pass | c) If something passes, it is a student |
| b) Every entity that is a student passes | d) There exists at least one student who passes |

Q5. Unification in FOL is the process of finding substitutions that:

- | | |
|---|---|
| a) Convert clauses to CNF | c) Prove a goal using backward chaining |
| b) Make two logical expressions syntactically identical | d) Derive facts using Modus Ponens |

Q6. In the Resolution algorithm, deriving the empty clause $\square$ indicates:

- | | |
|-------------------------------------|--|
| a) The knowledge base is consistent | c) A contradiction — the original goal is proven |
| b) The negated goal is satisfiable | d) More clauses need to be added |

Q7. Forward Chaining is best described as a _____ strategy.

- | | |
|----------------------------|--------------------------------|
| a) Goal-driven / top-down | c) Heuristic-based depth-first |
| b) Data-driven / bottom-up | d) Probabilistic inference |

Q8. Backward Chaining begins reasoning from:

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|---|--|
| a) All known facts toward a conclusion | c) A random clause in the knowledge base |
| b) The goal and works backward to verify supporting facts | d) The resolution refutation tree |

Q9. Which step is NOT part of converting a sentence to CNF?

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|--|--|
| a) <i>Eliminate biconditionals</i> ($\leftrightarrow$) | c) Apply Skolemization to remove existential quantifiers |
| b) Move negations inward using De Morgan's laws | d) Distribute AND over OR |

Q10. Knowledge Engineering in FOL involves:

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|--|---|
| a) Training a neural network on domain-specific data | c) Writing heuristic search strategies |
| b) Defining domain concepts, relations, and axioms formally in FOL | d) Encoding reward functions for reinforcement learning |

Q11. *Modus Ponens states: If $P \rightarrow Q$ is known and P is true, then:*

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|---------------------|---------------------------|
| a) $\neg Q$ is true | c) P is uncertain |
| b) Q must be true | d) The KB is inconsistent |

Q12. The Most General Unifier (MGU) is the substitution that:

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|--|---|
| a) Is the most specific possible match | c) Eliminates all variables from a clause |
| b) Makes two terms identical with minimal variable binding | d) Applies resolution to produce CNF |

Q13. Which inference method is more appropriate when a specific goal is known?

- | | |
|-----------------------|----------------------|
| a) Forward Chaining | c) Backward Chaining |
| b) Truth Table method | d) A* search |

Q14. Propositional Logic differs from First Order Logic because it:

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|---|-------------------------------|
| a) Supports probabilistic reasoning | c) Is always more expressive |
| b) Cannot represent objects, relations, or quantifiers — only truth values of atomic propositions | d) Handles uncertainty better |

Q15. Resolution Refutation proves a goal G by:

- | | |
|---|--------------------------------------|
| a) Directly deriving G from the KB using forward chaining | c) Checking G in the truth table |
| b) Adding $\neg G$ to the KB and deriving a contradiction (empty clause $\square$) | d) Applying Modus Tollens repeatedly |

Q16. Learning from observation in AI refers to:

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|--|---|
| a) An agent receiving numerical rewards from the environment | c) Acquiring knowledge by following explicit symbolic rules |
|--|---|

- | | |
|---|--|
| b) Improving performance by examining examples or experience from the environment | d) Memorizing a fixed dataset without generalization |
|---|--|

Q17. Inductive learning is best described as:

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|--|--|
| a) Deriving specific conclusions from general rules | c) Memorizing training data exactly |
| b) Generalizing a hypothesis from specific training examples | d) Applying search algorithms to find optimal policies |

Q18. In a Decision Tree, what does each internal node represent?

- | | |
|---|-------------------------------------|
| a) A class label / output decision | c) A test on an attribute / feature |
| b) A probability distribution over outcomes | d) A training data sample |

Q19. Explanation-Based Learning (EBL) differs from inductive learning because it:

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|--|---|
| a) Requires large amounts of training data to generalize | c) Uses prior domain knowledge to explain and generalize a single example |
| b) Learns only from numeric data using regression | d) Applies unsupervised clustering to find hidden patterns |

Q20. In statistical learning, a Naive Bayes classifier assumes:

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|---|---|
| a) All features are conditionally dependent on each other given the class | c) All features are conditionally independent given the class label |
| b) The data must follow a uniform distribution | d) Labels are determined by a reward function |

Q21. Learning with complete data (no hidden variables) in a Bayesian network involves:

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|---|---|
| a) Using the EM algorithm to estimate missing values | c) Directly computing maximum likelihood estimates from observed data |
| b) Training a deep neural network with dropout regularization | d) Applying reinforcement signals to update weights |

Q22. The Expectation-Maximization (EM) algorithm is used when:

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|---|--|
| a) Training data is fully labeled and no variables are hidden | c) The dataset contains hidden (latent) variables that cannot be directly observed |
| b) An agent explores an unknown environment using trial and error | d) Backpropagation fails to converge in a neural network |

Q23. In a Neural Network, the activation function serves to:

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|--|---|
| a) Initialize the weights of each neuron to zero | c) Introduce non-linearity to enable learning of complex patterns |
| b) Normalize the input dataset before training | d) Store previously learned examples in memory |

Q24. Which of the following correctly describes backpropagation in a neural network?

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|--|---|
| a) Weights are updated in the forward pass using input gradients | c) The error is propagated backward from output to input layers to adjust weights |
| b) Outputs of each layer are stored and reused in later epochs | d) A separate model is trained for each hidden layer independently |

Q25. A Support Vector Machine (SVM) — a type of kernel machine — classifies data by:

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|---|---|
| a) Minimizing the number of training examples used | c) Finding the maximum-margin hyperplane that separates classes |
| b) Grouping similar data points into clusters using centroids | d) Updating weights using Q-values from an environment |

Q26. The kernel trick in kernel machines (SVMs) allows:

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|--|---|
| a) Reducing the number of support vectors needed | c) Operating in a high-dimensional feature space without explicitly computing coordinates |
| b) Accelerating gradient descent convergence | d) Replacing hidden layers in a deep neural network |

Q27. In Reinforcement Learning, the agent aims to maximize:

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|---|--|
| a) Classification accuracy on a fixed test set | c) Cumulative reward received over time through interaction with the environment |
| b) The number of features selected from the training data | d) The size of the neural network hidden layer |

Q28. The Q-Learning algorithm in Reinforcement Learning is classified as:

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|---|---|
| a) A supervised learning algorithm requiring labeled datasets | c) A model-free, off-policy temporal difference learning method |
| b) An unsupervised clustering technique using centroids | d) A decision-tree-based ensemble classifier |

Q29. Overfitting in machine learning occurs when:

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|---|--|
| a) The model performs poorly on both training and test data | c) The model fits training data too closely and generalizes poorly to new data |
| b) The learning rate is set too low during gradient descent | d) The training set has more features than training examples |

Q30. Which of the following best distinguishes Reinforcement Learning from Supervised Learning?

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|---|--|
| a) RL uses labeled input-output pairs; SL uses reward signals | c) RL learns via trial-and-error with reward feedback; SL learns from labeled examples |
|---|--|

b) Both RL and SL require explicit reward functions d) RL is always faster to converge than SL on any dataset

- Q1 — b) An agent that uses a knowledge base and logical inference to decide actions
 Q2 — b) $\neg P \vee Q$
 Q3 — b) A conjunction of disjunctions (clauses)
 Q4 — b) Every entity that is a student passes
 Q5 — b) Make two logical expressions syntactically identical
 Q6 — c) A contradiction — the original goal is proven
 Q7 — b) Data-driven / bottom-up
 Q8 — b) The goal and works backward to verify supporting facts
 Q9 — c) Apply Skolemization to remove existential quantifiers
 Q10 — b) Defining domain concepts, relations, and axioms formally in FOL
 Q11 — b) Q must be true
 Q12 — b) Makes two terms identical with minimal variable binding
 Q13 — c) Backward Chaining
 Q14 — b) Cannot represent objects, relations, or quantifiers — only truth values of atomic propositions
 Q15 — b) Adding $\neg G$ to the KB and deriving a contradiction (empty clause $\square$)
 Q16 — b) Improving performance by examining examples or experience from the environment
 Q17 — b) Generalizing a hypothesis from specific training examples
 Q18 — c) A test on an attribute / feature
 Q19 — c) Uses prior domain knowledge to explain and generalize a single example
 Q20 — c) All features are conditionally independent given the class label
 Q21 — c) Directly computing maximum likelihood estimates from observed data
 Q22 — c) The dataset contains hidden (latent) variables that cannot be directly observed
 Q23 — c) Introduce non-linearity to enable learning of complex patterns
 Q24 — c) The error is propagated backward from output to input layers to adjust weights
 Q25 — c) Finding the maximum-margin hyperplane that separates classes
 Q26 — c) Operating in a high-dimensional feature space without explicitly computing coordinates
 Q27 — c) Cumulative reward received over time through interaction with the environment
 Q28 — c) A model-free, off-policy temporal difference learning method
 Q29 — c) The model fits training data too closely and generalizes poorly to new data
 Q30 — c) RL learns via trial-and-error with reward feedback; SL learns from labeled examples

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect

Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions